process, O_2 the ultimate acceptor of electrons gets reduced to water.

12.11.2 Anaerobic respiration

Anaerobic respiration takes place without oxygen. Glucose is converted into ethanol (Ethanol fermentation by yeast) or lactic acid (lactic acid fermentation by bacteria).

$$C_6H_{12}O_6 \rightarrow 2CO_2 + 2C_2H_5OH + \text{Energy (ATP)}$$

12.11.3 Respiratory quotient (R.Q)

Respiratory quotient is the ratio of volume of carbon dioxide liberated and the volume of oxygen consumed during respiration. It is expressed as

$$RQ = \frac{\text{Volume of } CO_2 \text{ liberated}}{\text{Volume of } O_2 \text{ consumed}}$$

Points to Remember

- Tissue is a group of similar or dissimilar cells, having a common orgin and performing similar functions.
- Plants are capable of synthesizing glucose from CO_2 and H_2O in the presence of light, by the process of photosynthesis.
- Light reaction takes place in grana of chloroplast.
- Dark reaction takes place in stroma of chloroplast.
- Respiration involves both external and cellular respiration.
- Aerobic respiration takes place in the presence of oxygen.
- Aerobic respiration occurs in three major steps like Glycolysis, Krebs cycle and Electron transport chain.

TEXTBOOK EVALUATION

JZKQCK

I. Choose the correct answer

1. Casparian strips are present in the _____ of the root.
   a) cortex b) pith
   c) pericycle d) endodermis
2. The endarch condition is the characteristic feature of
   a) root b) stem
   c) leaves d) flower
3. The xylem and phloem arranged side by side on same radius is called _____
   a) radial b) amphivasal
   c) conjoint d) None of these
4. Which is formed during anaerobic respiration
   a) Carbohydrate b) Ethyl alcohol
   b) Acetyl CoA d) Pyruvate
5. Kreb's cycle takes place in
   a) chloroplast
   b) mitochondrial matrix
   c) stomata
   d) inner mitochondrial membrane
6. Oxygen is produced at what point during photosynthesis ?
   a) when ATP is converted to ADP
   b) when CO_2 is fixed
   c) when H_2O is splitted
   d) All of these

II. Fill in the blanks.

1. The innermost layer of cortex in root is called _____.
2. Xylem and phloem are arranged in an alternate radii constitute a vascular bundle called _____.

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